

Yasin Rezvani

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EDUCATION

Shahrood University of Technology

B.Sc. in Computer Engineering

Shahrood, Iran

Sep 2022 – Sep 2027

- GPA: **17.04**/20
- Major GPA: **17.96**/20
- Thesis: “Brain Tumor MRI Dataset and Joint Deep Neural Network for Segmentation and Classification”

RESEARCH INTERESTS

Medical Image Analysis, Image Segmentation and Classification, Vision Transformers, Multi-Task Learning

PUBLICATIONS

- A. Fateh, **Y. Rezvani**, S. Rezvani, M. Fateh, V. Abolghasemi, “Swin-HAFNet: A Hierarchical Multi-Task Transformer for Brain Tumor Segmentation and Classification”, *Array*, under review, Jun. 2026.
- A. Fateh, **Y. Rezvani**, S. Moayedi, S. Rezvani, F. Fateh, M. Fateh, V. Abolghasemi, “BRISC: Annotated Dataset for Brain Tumor Segmentation and Classification”, *Scientific Data*, Feb. 2026.
- S. Rezvani, F. S. Siahkar, **Y. Rezvani**, A. A. Gharahbagh, V. Abolghasemi, “Single Image Denoising via a New Lightweight Learning-Based Model”, *IEEE Access*, Aug. 2024.
- **Y. Rezvani**, A. Rezvani, “The Serious Water Crisis and Our Responsibility Against It”, *Third National Conference on Modern Approaches to Education and Research*, Mar. 2019.

RESEARCH EXPERIENCE

Research Assistant

CVLab SHUT

Shahrood, Iran

Jan 2024 – Present

- Co-developed the BRISC dataset and Swin-HAFNet for brain tumor segmentation and classification.
- Contributed to data annotation, model training, and experimental evaluation.

Computer Vision Research Intern

Shahaab Co.

Shahrood, Iran

Jun 2024 – Sep 2024

- Developed YOLO-based models for Iranian vehicle classification and Persian license plate recognition.
- Contributed to dataset preparation, annotation, and model optimization.

ACADEMIC SERVICE

Peer Reviewer

Scientific Data

Sep 2025 – Present

- Reviewed a manuscript focused on scientific dataset creation, data collection, annotation, and preprocessing.

Peer Reviewer

IEEE Access

Jun 2024 – Present

- Reviewed 14 manuscripts in computer vision, medical image analysis, and image processing.

Speaker

TED-Ed Student Talks

Jun 2019 – Sep 2019

- Presented *U-Health*, an AI-driven platform to improve remote access to medical services for all communities.

PROJECTS

LL(1) Parser with Panic Mode Recovery

Supervisor: *Dr. Elias Khajeh Karimi* — Course: *Principles of Compiler Design*

Feb 2025 – Jun 2025

- Designed an interactive LL(1) parser featuring real-time parse tree visualization and Panic Mode error recovery.

Binary Search Tree (BST) Visualizer

Supervisor: *Dr. Maryam Khodabakhsh* — Course: *Data Structure*

Oct 2024 – Jan 2025

- Built a Java Swing application for interactive visualization of BST insertion and deletion.

Image Denoising Using FFT and DnCNN

Supervisor: *Dr. Esmaeel Tahanian* — Course: *Engineering Mathematics*

Feb 2024 – Jun 2024

- Evaluated FFT-based filtering and DnCNN for image denoising through comparative performance analysis.

Hands-On Machine Learning and Deep Learning with Python

Supervisor: *Dr. Mansoor Fateh* — Organization: *CVLab SHUT*

Jan 2024 – Apr 2024

- Explored machine learning and deep learning techniques through PyTorch and Keras implementations on the MNIST and CIFAR-10 datasets.

SKILLS

Programming Languages: Python, C++, Java

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, Ultralytics, NumPy, Pandas, Matplotlib, SciPy

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, Kaggle, Docker

CERTIFICATES

- **Elements of AI** – University of Helsinki, Jul 2026
- **Research Collaborations Module** – Elsevier, Jun 2025
- **Prompt Engineering for Everyone** – Cognitive Class, Feb 2024
- **Supervised Machine Learning: Regression and Classification** – Coursera, Feb 2024
- **Machine Learning with Python** – Cognitive Class, Feb 2024
- **Learning Python** – LinkedIn, Aug 2022

LANGUAGE PROFICIENCY

- **Persian:** Native
- **English:** Registered for the TOEFL iBT exam (scheduled for October 3, 2026).

REFERENCES

Mansoor Fateh

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Shahrood University of Technology, Shahrood, Iran

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Vahid Abolghasemi

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University of Essex, Colchester, United Kingdom

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Esmaeel Tahanian

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Shahrood University of Technology, Shahrood, Iran

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